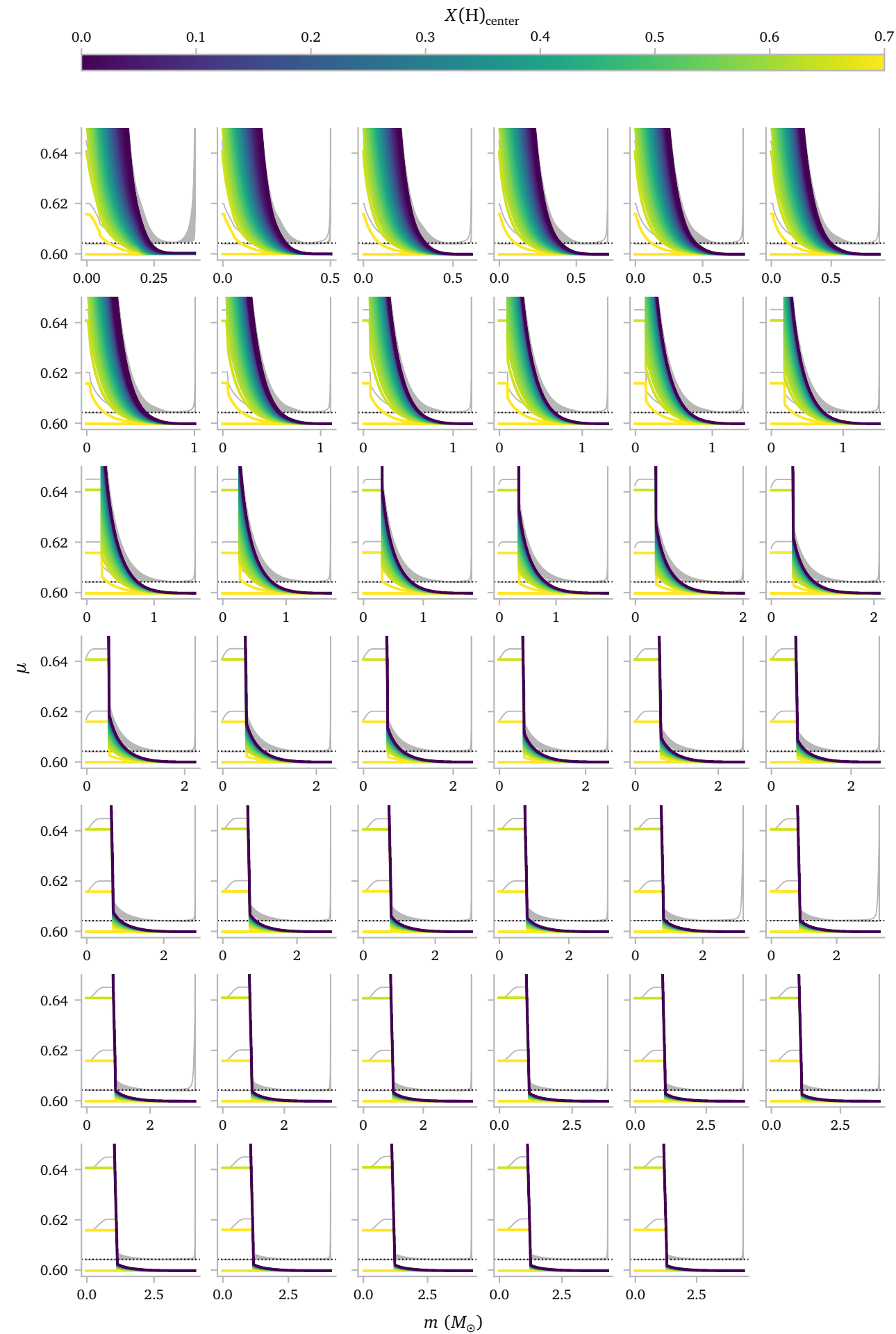


Notes Week 31

a. Fully Ionized Mean Molecular Weight

a.1 Main Sequence

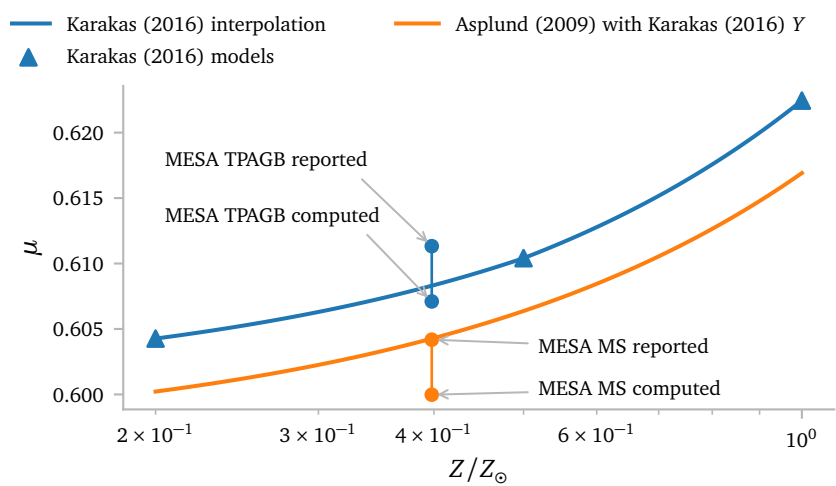
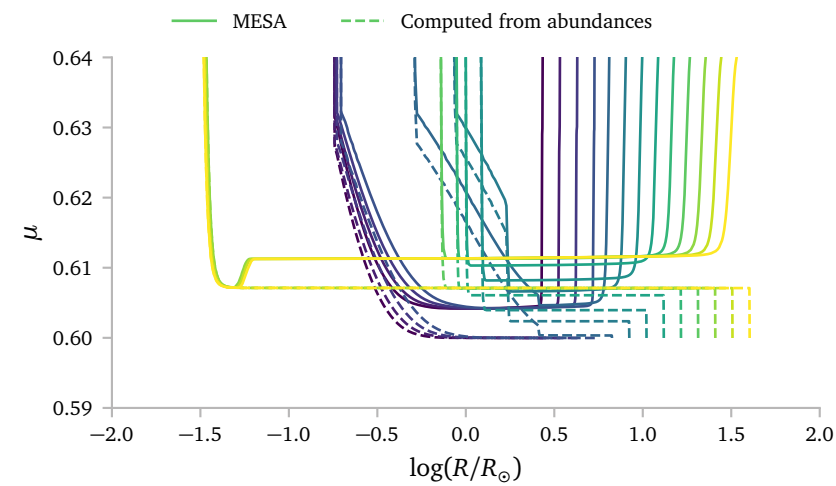
Since we are using the fully ionized mean molecular weights for the accreted material, it is only fair to compare this with the fully ionized mean molecular weights of MESA abundances.



The gray lines are the μ values as reported by *MESA*. The colored lines are the computed values from the abundances and the horizontal line is μ from [Asplund et al. \(2009\)](#). I *really* do not understand why the computed mean molecular weights from the *MESA* models are lower than reported μ from *MESA* and the computed [Asplund et al. \(2009\)](#) μ values.

I feel like the μ values reported from *MESA* are more representative because they agree with [Asplund et al. \(2009\)](#)(?)

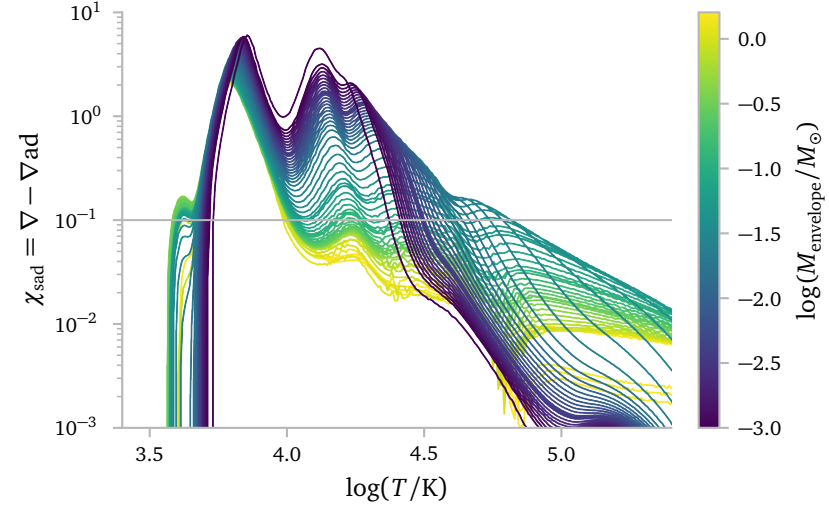
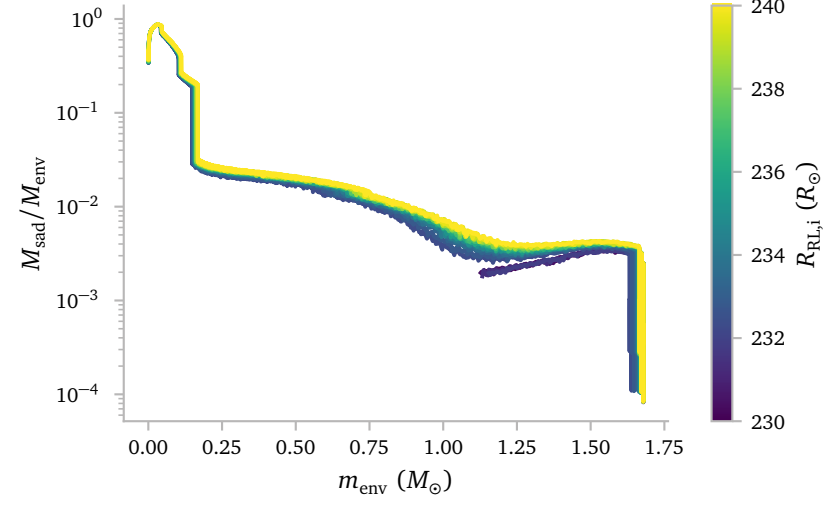
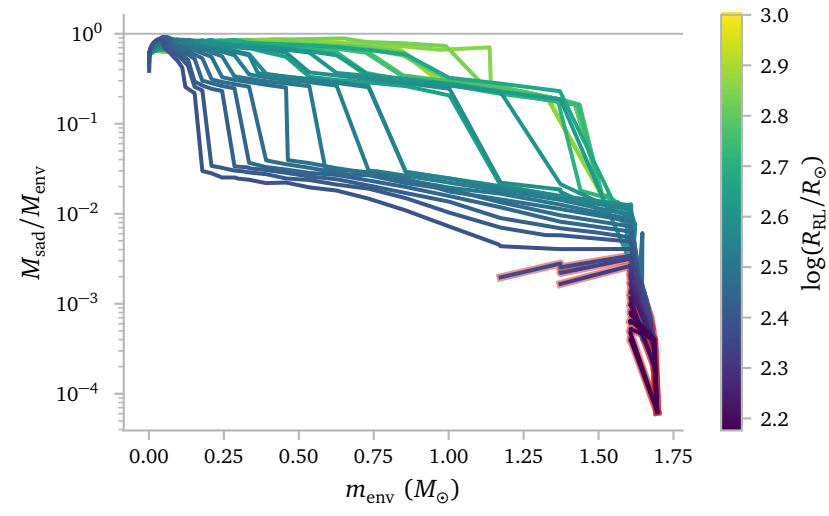
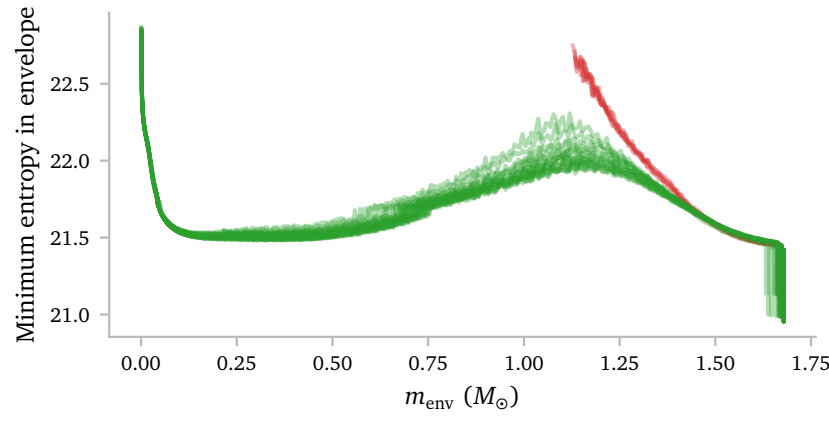
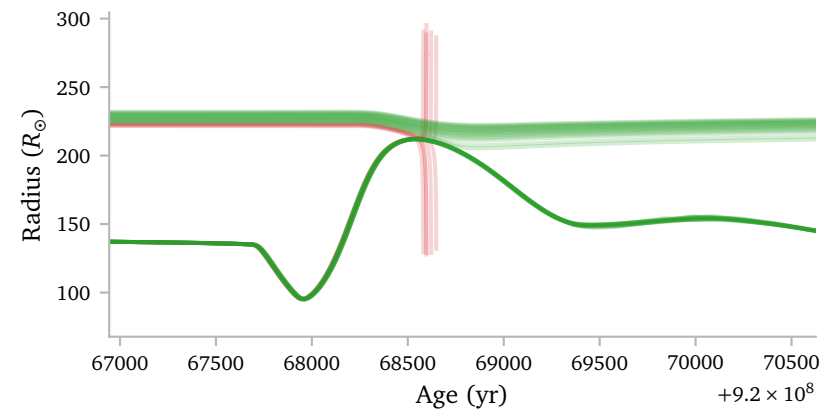
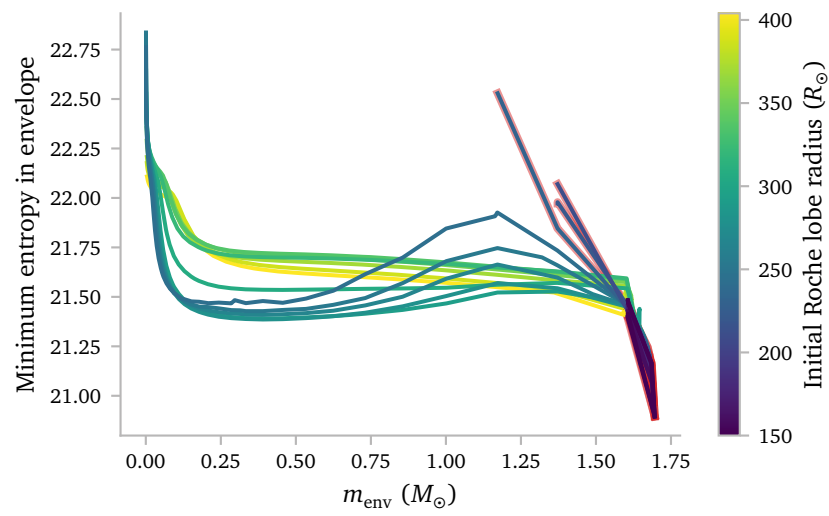
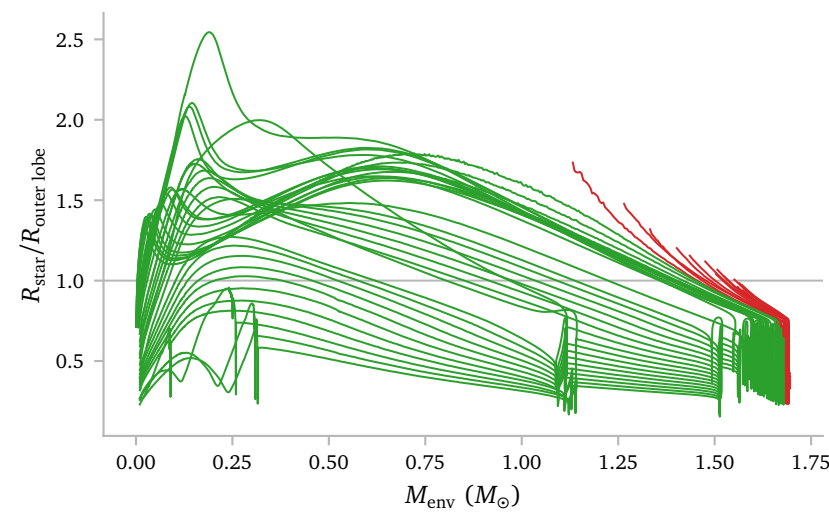
a.2 Giant branch and Thermally Pulsing Asymptotic Giant Branch



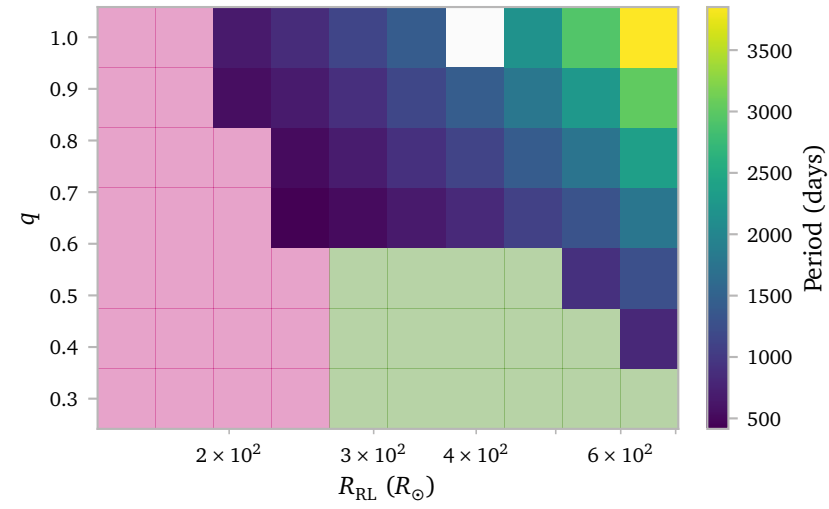
The mean molecular weights reported by the *MESA* models on the main sequence agree very well with the mean molecular weights of the [Asplund et al. \(2009\)](#).

On the TPAGB, it is exactly opposite. Here, the computed values from the abundances agrees much better with the μ computed from the [Karakas & Lugaro \(2016\)](#) abundances.

b. Failing models II

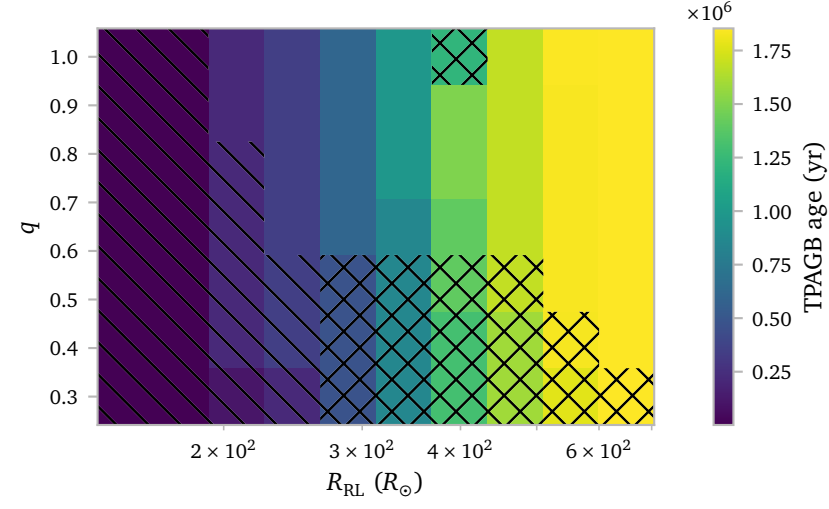


I want to return to this plot I made a while ago:

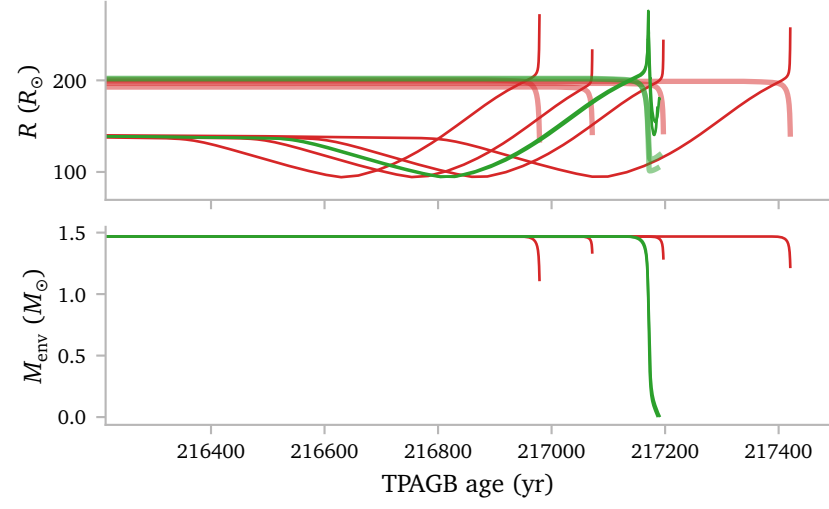


This is a grid with a $M = 2.0 M_{\odot}$ TPAGB-star, $\gamma = 1$ and $\delta = 0.5$. The pink squares are models that fail due to convergence problems. The green models end because the models spiral in.

I am curious if the models that fail, all happen in during the same thermal pulses. If however, it is possible to survive RLOF during a thermal pulse with a certain q , but not with another q .

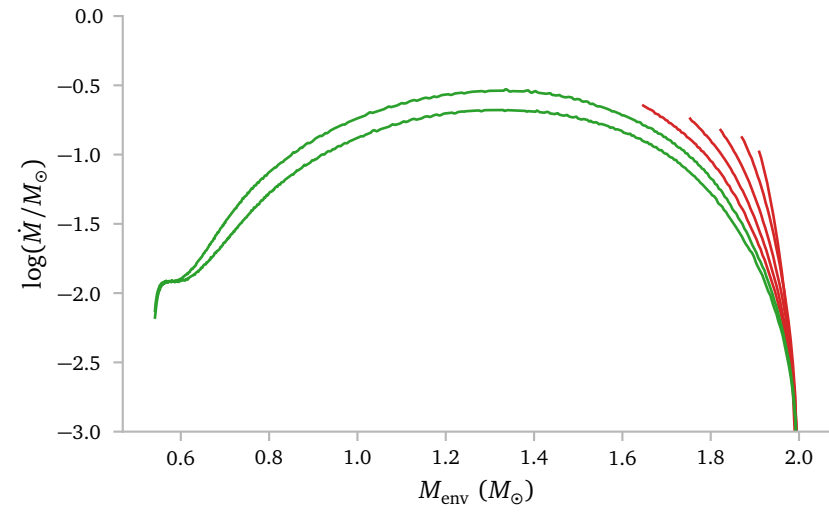


Clearly there are models that interact at roughly the same time (same color), but some of them crash, (dashed) while others do not.

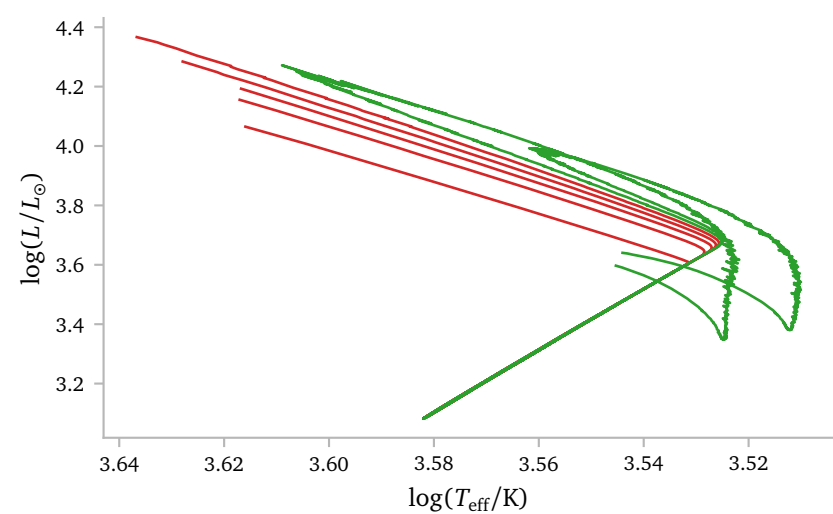


Here, I explicitly show this. The models interact during the same pulse, however, only the green lines, with the largest q , survive.

I guess the mass loss rate is another important consideration.



I show that here. The models with the largest q have the smallest $\dot{M}$ and thus strip the star slower. The differences here might seem minor, but this is a log scale so differences are actually quite large.

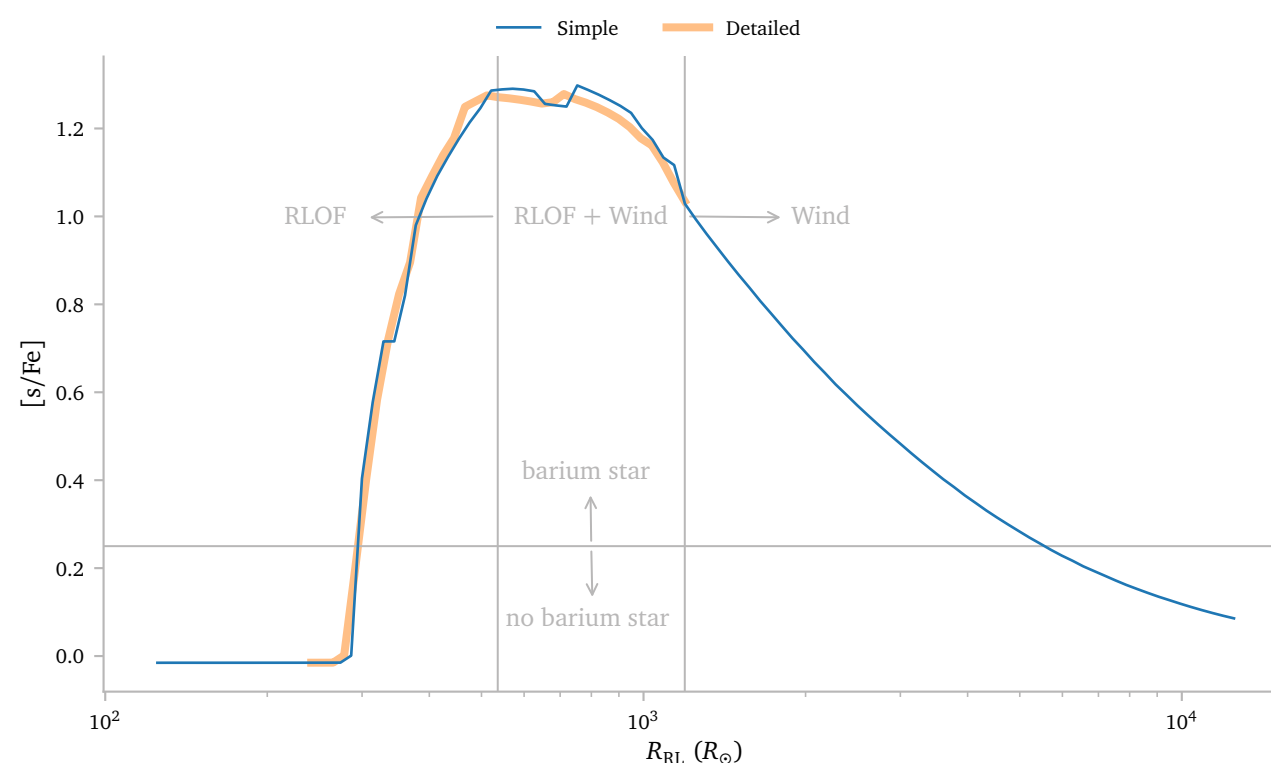


c. Grids

Using a very simple assumption, it is possible to quite accurately estimate the final accretor abundances of the detailed *MESA* simulations using only the simple direct binary evolution in python with *evolve.py*. The idea is this:

Follow the simple binary evolution as predicted by *evolve.py* exactly up to the point of RLOF. Once RLOF happens, assume that *all* of the remaining envelope mass is expelled instantly and accreted with a certain efficiency. To match *MESA* models, we should let ϵ_{MT} be this efficiency parameter. We then say that this material is accreted instantly at the last simple binary evolution step.

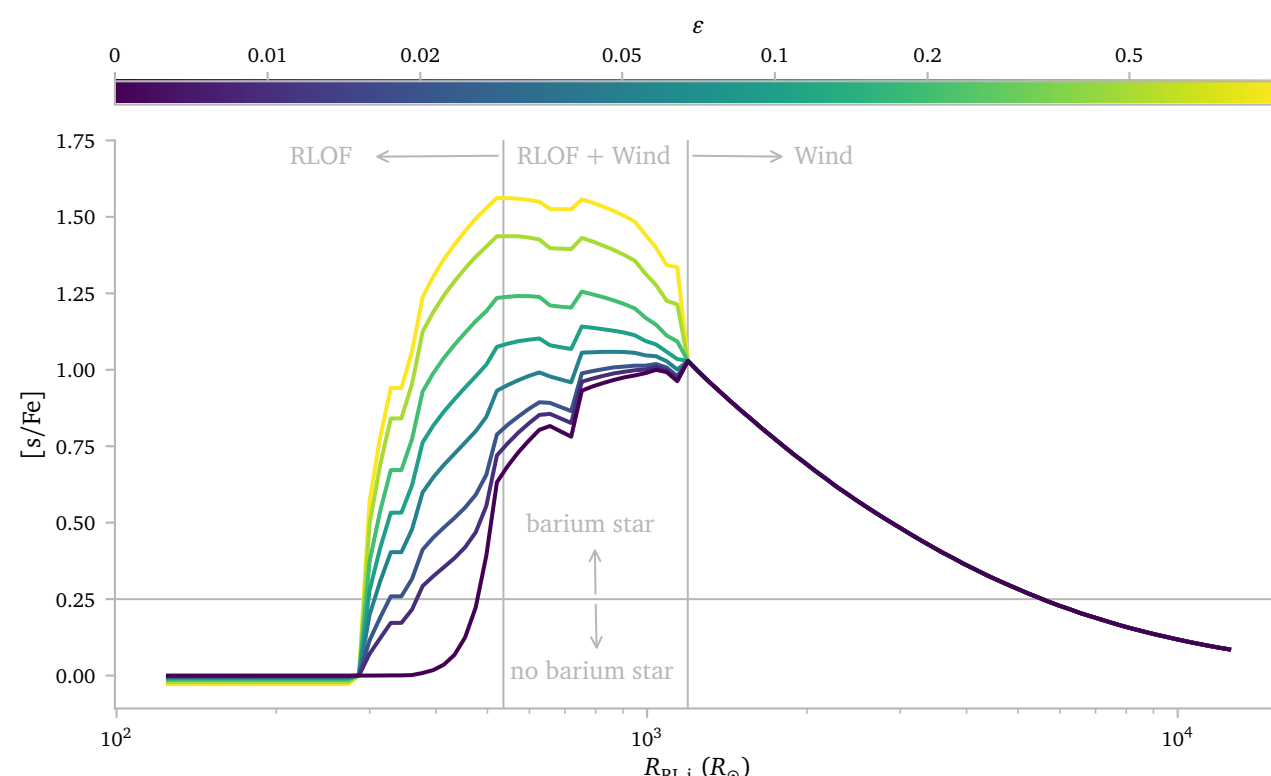
Below, I show this assumption and compare it to the R_{RL} grid that I simulated last week. These all have $M_{\text{TPGB},i} = 2.2 M_{\odot}$, $q = 0.6$ and $\epsilon = 0.25$.



Clearly, the two match very well, and the small differences are probably due to the final assumption that there is *no* wind during RLOF and the small differences between the orbital evolution of the detailed *MESA* models and the simple binary models.

It is also very easy to simulate large binaries that miss RLOF:

Below, I show the effects of the mass transfer efficiency on the final abundances.



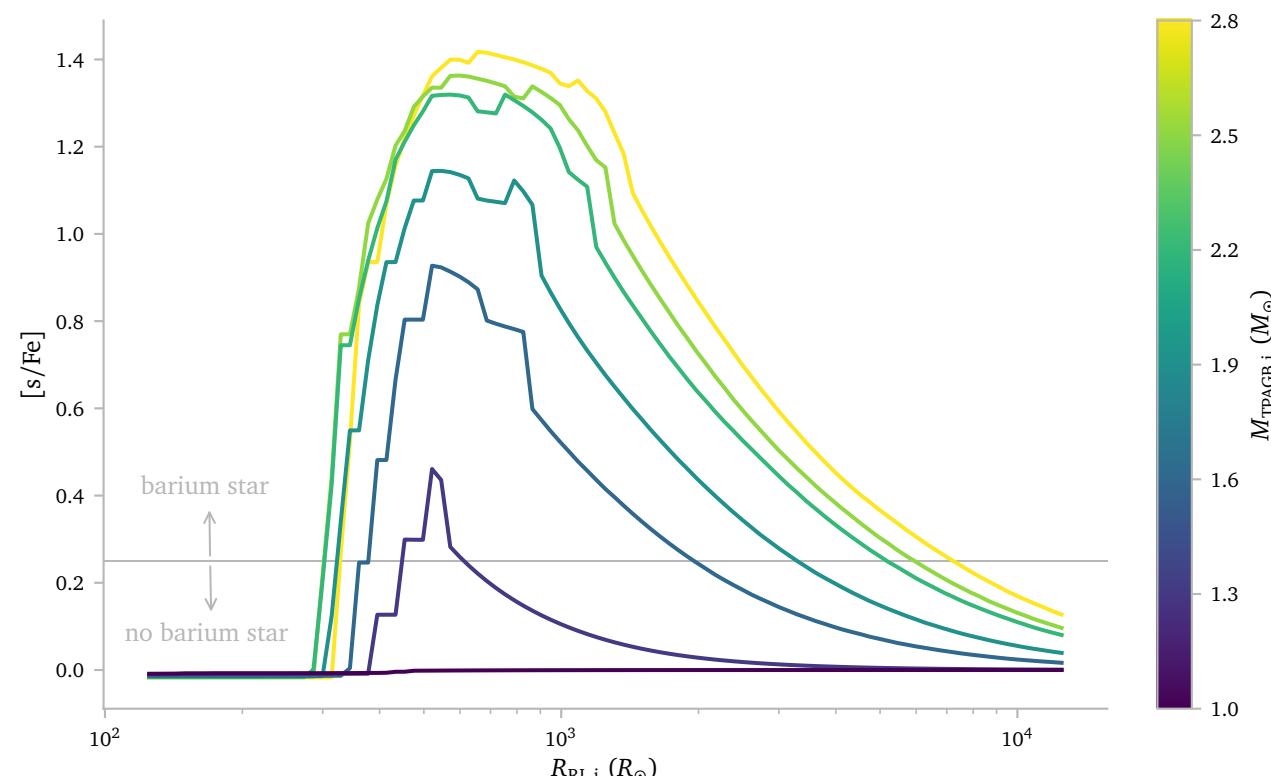
Interestingly, it seems like it is very easy to create barium stars even *without* any RLOF accretion. The enrichment for the purple line comes only from the wind accretion.

If we do have *RLOF* accretion, we can get higher abundances for smaller binaries. These binaries will also have smaller final periods. It also seems like we can create barium stars in very large binaries, just using wind accretion.

Lastly, we see a quite sharp drop between the models that undergo RLOF and those that do not. I *think* this is due to the envelope mass remaining during the last thermal pulse. If the star has, say $0.5 M_{\odot}$ of envelope remaining, this will be accreted with RLOF for models that *just* undergo RLOF, while this will *not* be accreted with the same efficiency for models that just barely miss RLOF.

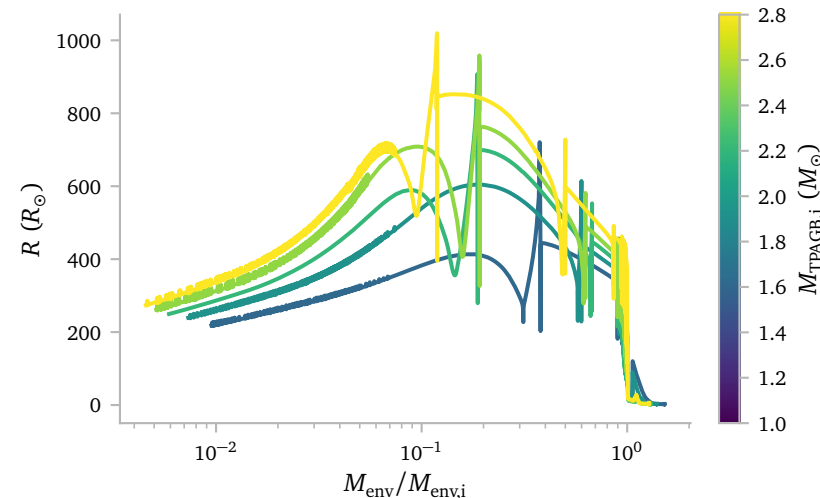
To test this, we can look at different masses, since this will shift the amount of envelope mass remaining during the last thermal pulse.

Here again, we have $q = 0.6$ and $\epsilon = 0.25$.



Here, it looks like the $1.6 M_{\odot}$ model has much more mass during the last thermal pulse than for example, the $2.8 M_{\odot}$ star.

I confirm this below.

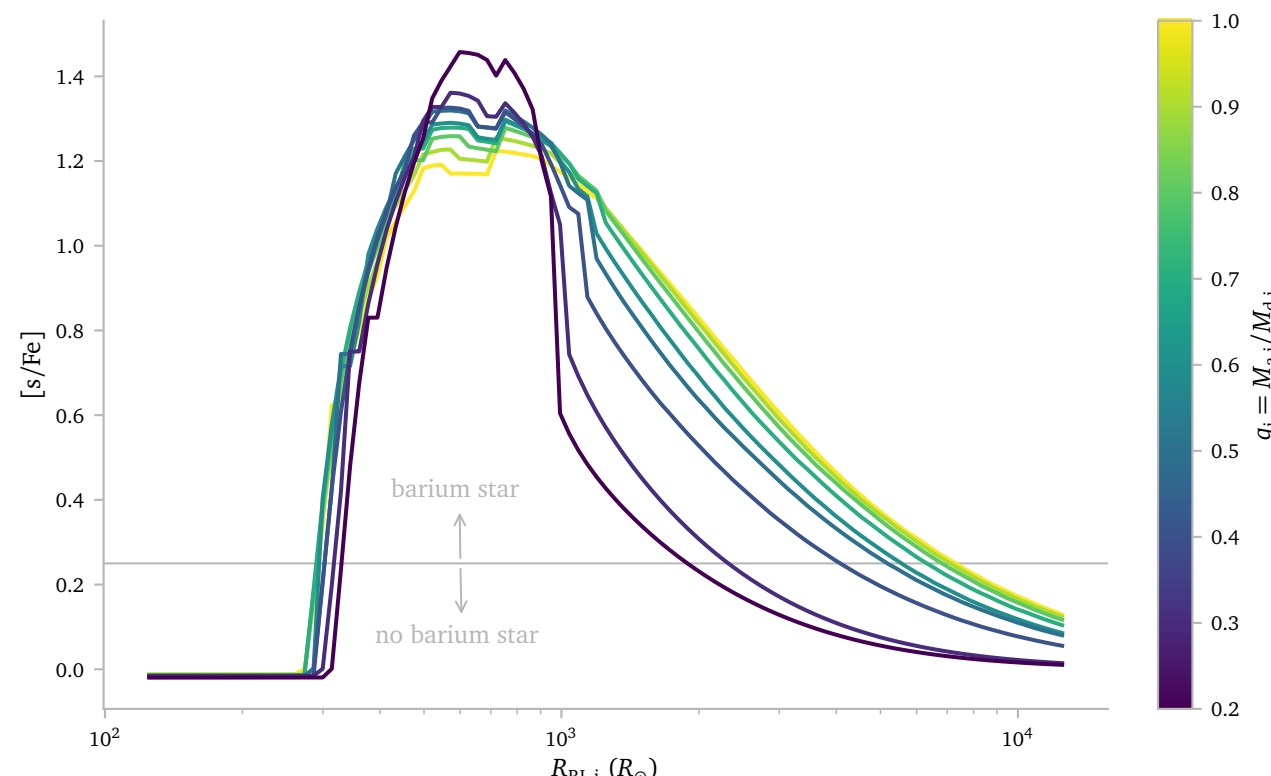


Clearly, when plotting the radius as a fraction of the envelope (and also, but less clearly, just as the envelope mass), we can see that the last thermal pulses for the various models happen at different remaining envelope fractions.

The $1.6 M_{\odot}$ model has relatively a lot more envelope mass remaining when the last pulse happens than the $2.8 M_{\odot}$ model. I think it then makes sense why we see the sharp drop when the models miss RLOF and why the height of this drop can depend quite strongly on the mass. It is really an effect of how much mass is remaining during the last TP.

We can also change the initial mass ratio q . This should have some interesting effects. In particular, the [Saladino et al. \(2018, 2019\)](#) wind prescription is dependent on q , q shifts the Roche lobe radius in a non-trivial way. A smaller q will mean that R_{RL}/a is larger, which means for the same initial R_{RL} , the initial separation a_i is larger.

Lastly, the RLOF mass transfer is depends on the initial mass ratio due to the fraction $\Delta M_{\text{acc}}/M_{\text{acc}}$. In this simple case, ΔM_{acc} is constant, while $M_{\text{acc}} = q \cdot M_{\text{TPGB}}$. So for smaller q , we expect larger abundances during RLOF.

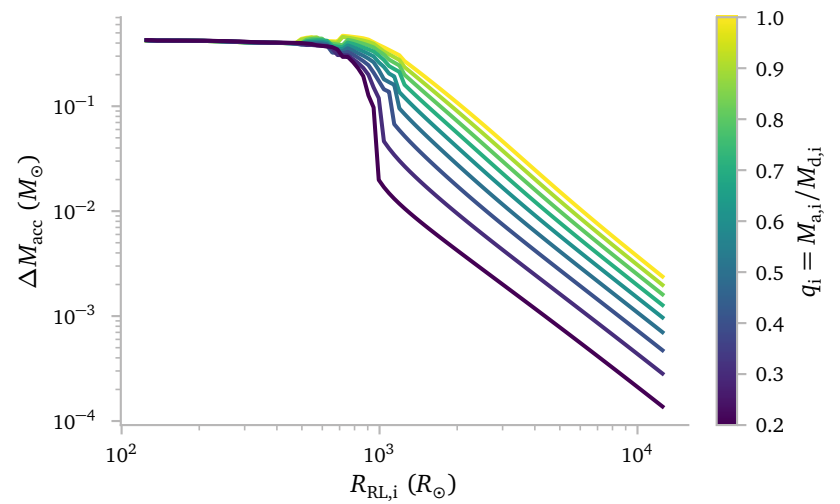


We indeed see that the dependence on q is quite important. In this simple case, a smaller q will lead to *higher* abundances during RLOF but *lower* abundances during the wind mass loss regime.

In reality, the effect is probably more nuanced. I would not at all be surprised if there is some effect of q on the mass transfer efficiency. Since models with lower q tend to experience much more extreme RLOF, with higher mass loss rates, it would not surprise me if more of this mass is lost, which would counteract the effect of higher abundances due to the lower accretor mass.

We also see that the discontinuity between the RLOF and the wind regime is much more pronounced for the lower q models. I think this is probably because the mass transfer efficiency of wind is lowered with lower q , while this is not the case during RLOF.

Let me explicitly test this. Below, I show the amount of accreted mass for the different q models.



We can indeed clearly see that the models with lower q experience much less wind accretion.

There also exists a nice powerlaw between the accreted material due to wind and the initial Roche lobe radius (or initial binary separation).

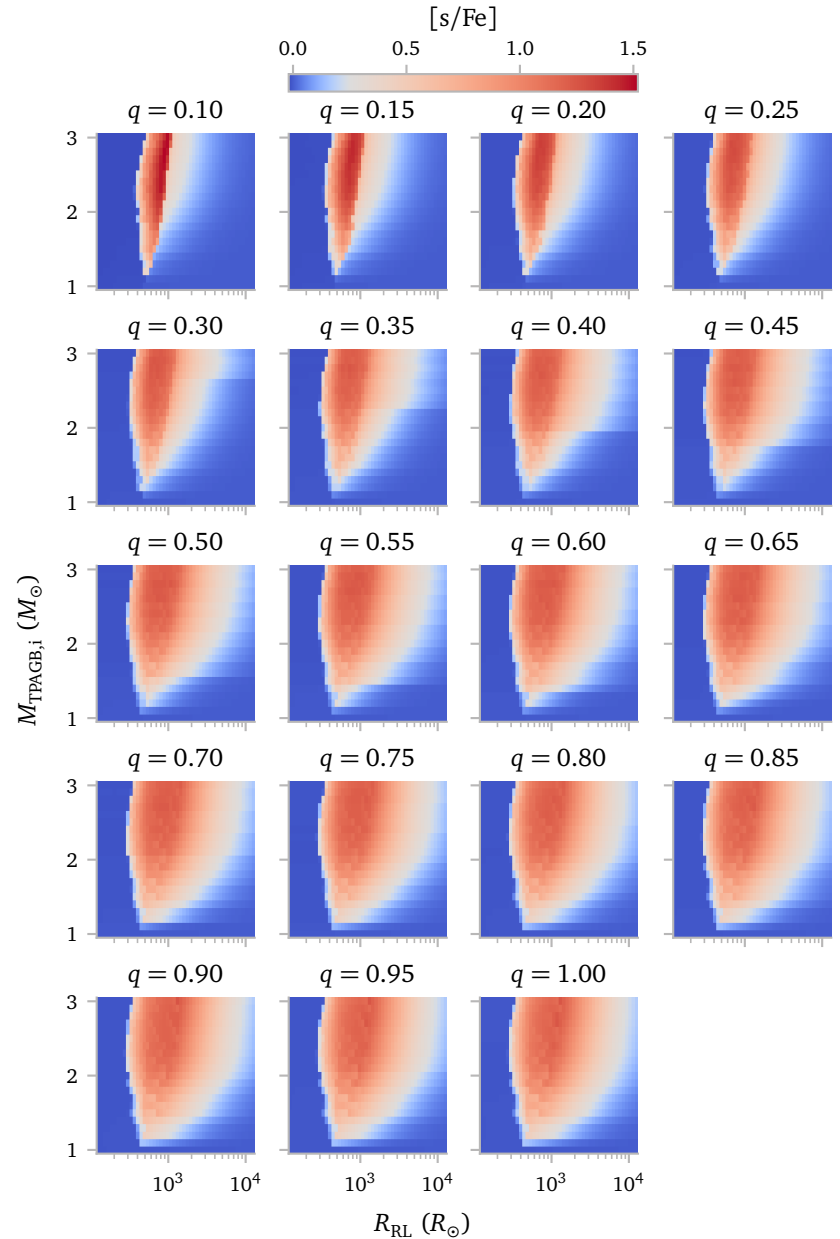
Since it is *much* cheaper to compute the abundances of binary models using this simple approach then using detailed *MESA* simulations, it makes sense to explore the parameter space using these simple models to get an *estimate* for the abundances of the accretor.

I decided to use the following parameters:

- 48 logspaced initial Roche lobe radii from $125 R_\odot$ to $1250 R_\odot$.
- 21 linearly spaced initial TPAGB masses from $1.0 M_\odot$ to $3.0 M_\odot$.
- 19 linearly spaced initial mass ratios from $q = 0.1$ to 1.0 .
- 20 logspaced RLOF accretion efficiencies from 0.001 to 1 .

In total, this gives a grid of 383040 models, which only took a couple of hours to simulate¹.

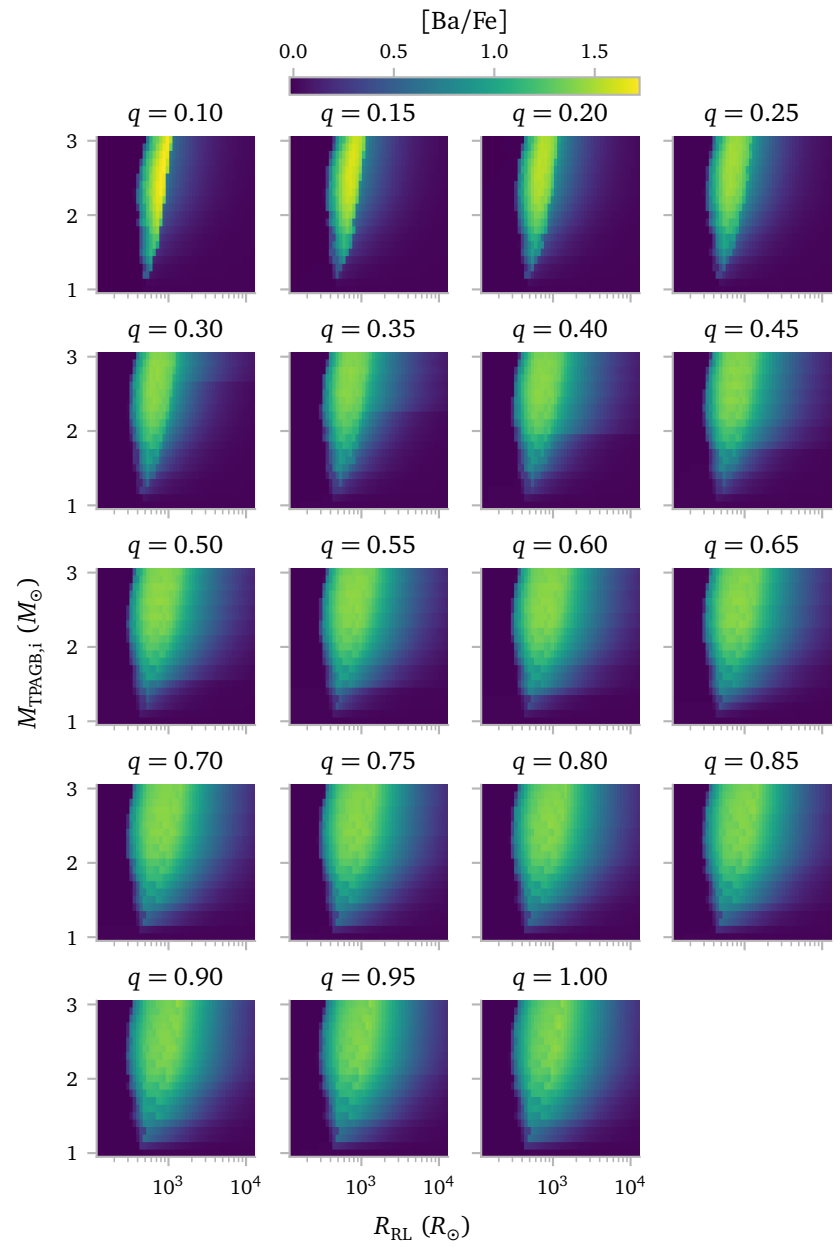
Obviously, because this is a 4D grid, it is impossible to show in its entirety at once. To start, let me fix ϵ to $\epsilon \approx 0.113$.



Fixed $\epsilon \approx 0.113$.

This grid shows the initial Roche lobe radius versus the initial TPAGB age in every panel. The different panels have different initial mass-ratios. The colors represent the expected final observed $[s/Fe]$ for the accretor. We can see clear differences between the different panels, but (as expected), the minimum mass for significant s-process enrichment is independent of q . It seems like we need at least a TPAGB star that is $1.3 M_\odot$ to get significantly large abundances. We then also see that the highest abundances occur in systems with the heaviest donors. This makes sense, these systems can donate more mass, *and* they have more dredge-up then systems with lighter TPAGB donors. Lastly, we see that the initial mass ratio changes the *width* of the enriched region quite significantly. This has two reasons:

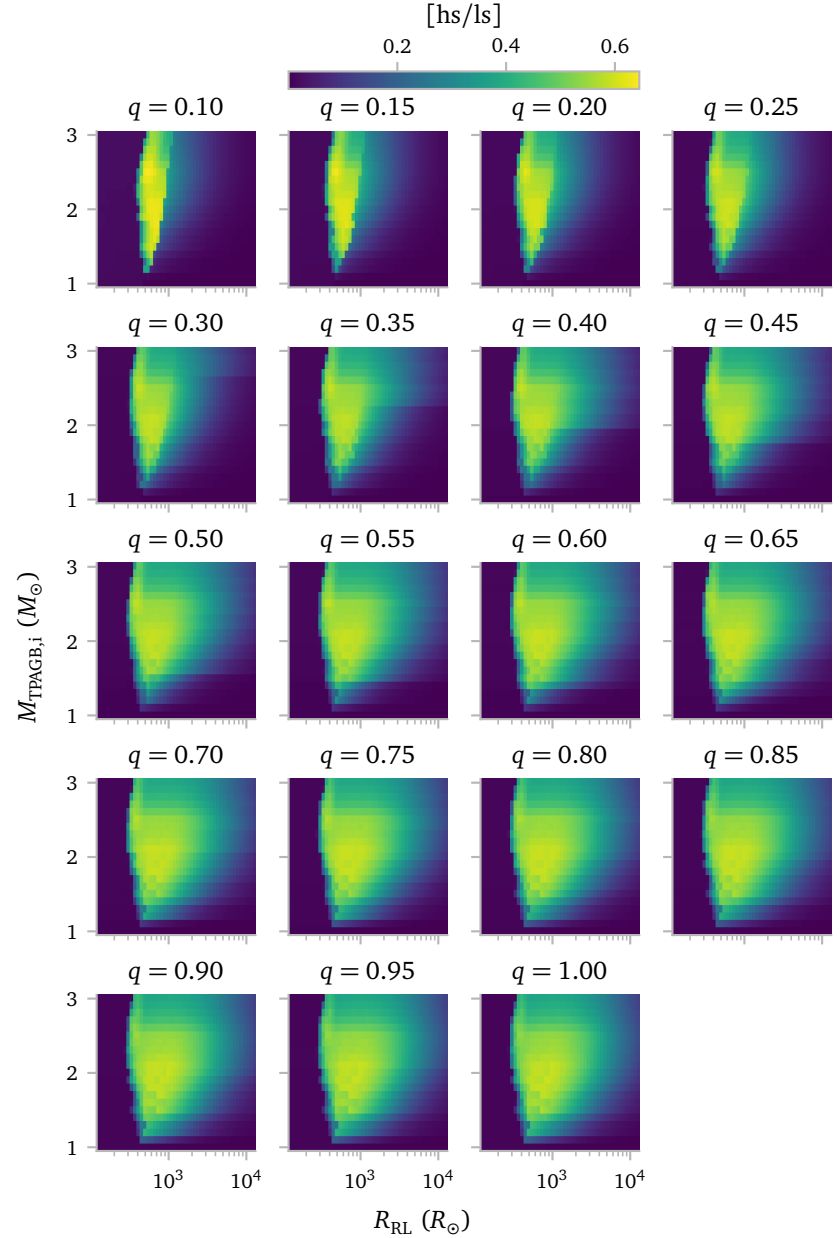
1. The smaller q leads to faster orbital shrinkage due to winds, which shifts the boundary between RLOF and wind-only MT to lower initial Roche lobe radii.
2. Wind mass transfer is much less efficient for low q than it is for high q . This means enrichment drops off much quicker in the wind-only regime for systems with small q then for systems with large q .



Fixed $\epsilon \approx 0.113$.

This grid shows the initial Roche lobe radius versus the initial TPAGB age in every panel. The different panels have different initial mass-ratios. The colors represent the expected final observed $[Ba/Fe]$ for the accretor.

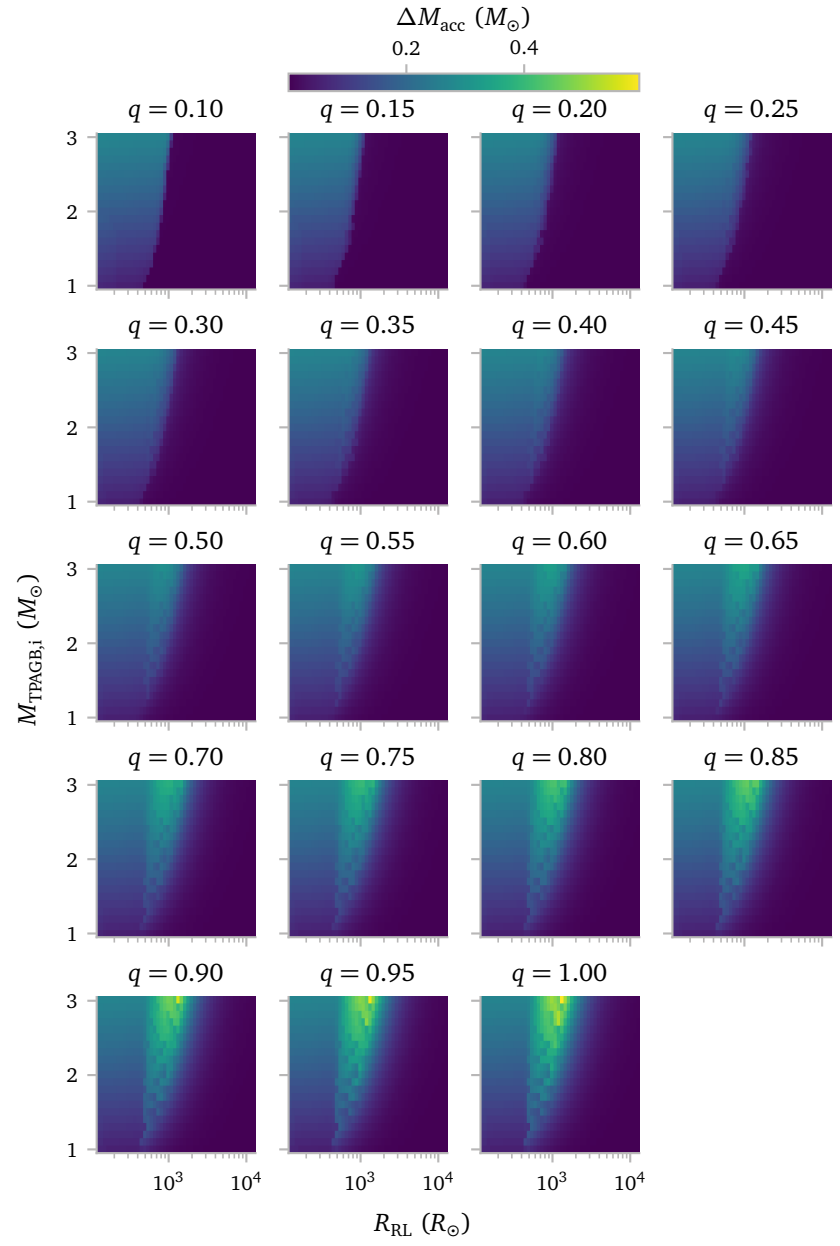
There is only a small difference between these plots and the ones for $[s/Fe]$. In general $[Ba/Fe]$ is a bit larger, only because of the 7 most important s-process elements, barium is slightly more abundant than average.



Fixed $\epsilon \approx 0.113$.

This grid shows the initial Roche lobe radius versus the initial TPAGB age in every panel. The different panels have different initial mass-ratios. The colors represent the expected final observed $[hs/ls]$ for the accretor.

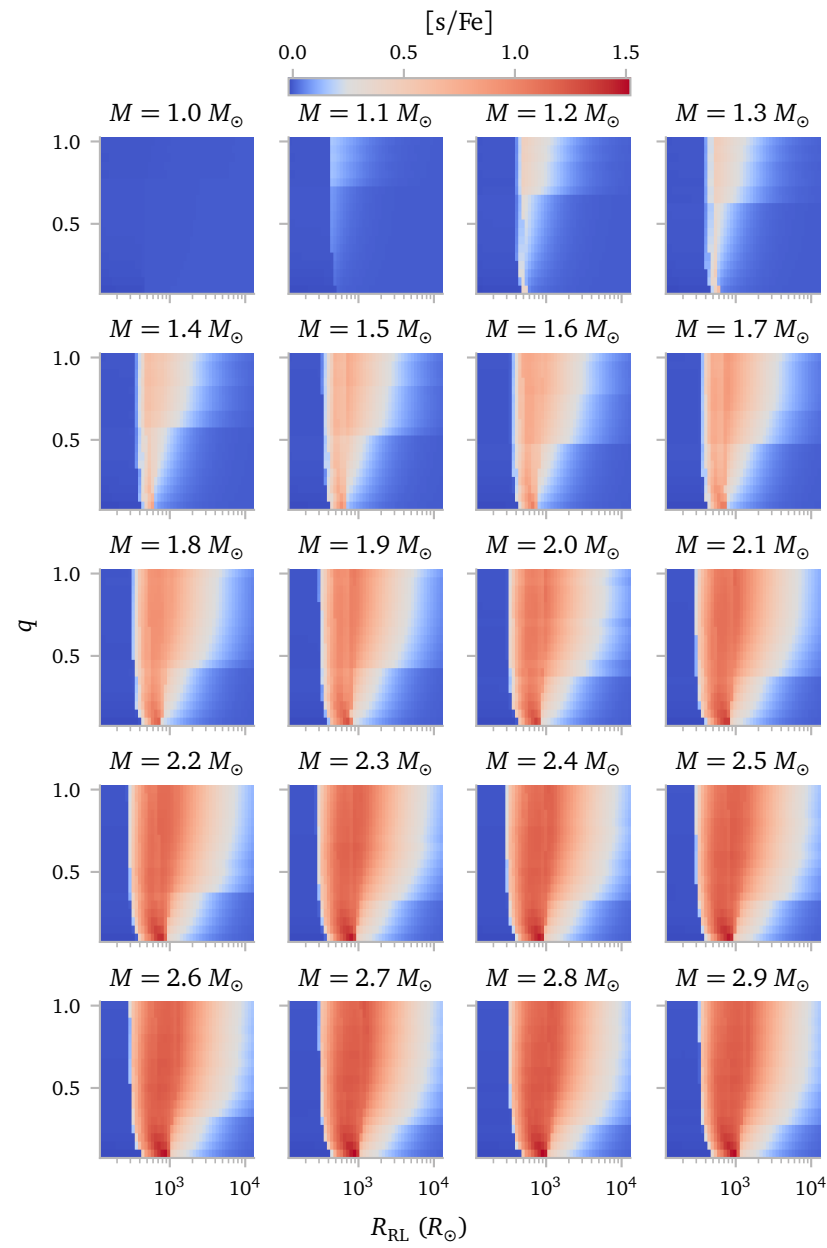
I do not think this plot is very important. It seems to mostly be linked with the general $[s/Fe]$ abundances. If I would have to say something about this, it *seems* like the get slightly lower $[hs/ls]$ for the most massive stars compared to the $[s/Fe]$ abundances, which do not show this decrease with mass. This is consistent with Karakas & Lugaro (2016, Figure 17) where $[hs/ls]$ rises to about 0.6 from $M = 1.3 M_\odot$ to $\sim 2.4 M_\odot$. Then it decreases to about 0.3 for $M = 3.0 M_\odot$. Now since I am interpolating between these models, and slightly lower metallicity ones, we expect roughly the same properties, although maybe slightly higher because $[hs/ls]$ seems to grow quite significantly with decreasing metallicity.



Fixed $\epsilon \approx 0.113$.

This grid shows the initial Roche lobe radius versus the initial TPAGB age in every panel. The different panels have different initial mass-ratios. The colors represent the expected final ΔM_{acc} for the accretor.

These panels very clearly show the boundary between the RLOF (+ wind) regime and the winds only regime. This is so visible because the winds, at least for the very large roche lobe radii, have a much lower accretion efficiency than the fixed value of 0.113 that I am plotting for RLOF. However, this is not the full picture. We can also see that for the larger q 's, there are 3 visible regimes. The lowest accretion regime is still the winds only regime, but the regime to the left of that now clearly shows the divide between systems that undergo significant amounts of wind before undergoing RLOF and systems that do not. The systems that *do* experience a lot of winds will generally have slightly *more* accreted material. At first, this might seem at odds with the lower accretion in the winds only regime, but it is actually not, because the accretion efficiency of the winds strongly depends on the size of the orbit. The much smaller orbits during this wind RLOF + wind regime can lead to *larger* wind accretion efficiencies than the fixed 0.113 for RLOF.



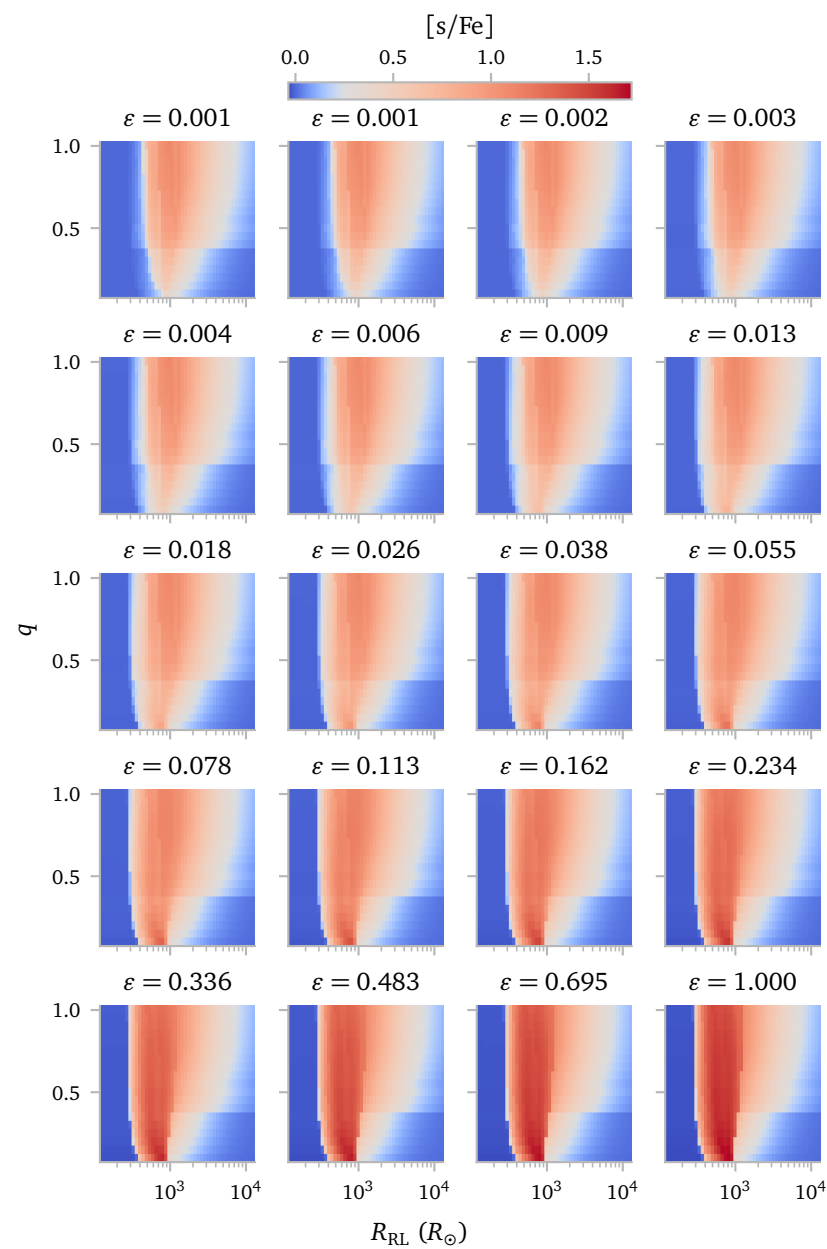
Fixed $\varepsilon \approx 0.113$.

This grid shows the initial Roche lobe radius versus the initial mass ratio q in every panel. The different panels have different initial TPAGB masses. The colors represent the expected final observed $[s/Fe]$ values.

One thing that is immediately clear is that the lower mass models, up to say $1.3 M_{\odot}$ can *not* or only under very specific circumstances create a barium star. It then also seems like once you reach a certain TPAGB mass, say something like $2.2 M_{\odot}$, the final s-process abundances of the accretor do not change a lot.

The increase in $[s/Fe]$ towards lower q in the RLOF regime is due to the lower accretor mass, while the decrease in width is due to the less efficient wind mass transfer. This creates the general South-America shaped regions.

Lastly, there are quite clear horizontal lines visible in the abundances at the wind only regime. They seem to shift to lower q for larger donor masses. It took me a *long* time to realise this is due to the “full mixing” that gets enabled for companion stars below $0.8 M_{\odot}$.



Fixed mass $M_{TPAGB,i} = 2.2 M_{\odot}$.

This grid shows the initial Roche lobe radius versus the initial mass ratio q in every panel. The different panels have different ε_{MT} values during RLOF. The color represents the final $[s/Fe]$ abundance.

References

- Asplund, M., Grevesse, N., Sauval, A. J., & Scott, P. (2009) *The Chemical Composition of the Sun*. [ARA&A47](#), 481.
- Karakas, A. I. & Lugaro, M. (2016) *Stellar Yields from Metal-rich Asymptotic Giant Branch Models*. [ApJ825](#), 26.
- Saladino, M. I., Pols, O. R., & Abate, C. (2019) *Slowly, slowly in the wind. 3D hydrodynamical simulations of wind mass transfer and angular-momentum loss in AGB binary systems*. [A&A626](#), A68.
- Saladino, M. I., Pols, O. R., van der Helm, E., Pelupessy, I., & Portegies Zwart, S. (2018) *Gone with the wind: the impact of wind mass transfer on the orbital evolution of AGB binary systems*. [A&A618](#), A50.